

Modelling, Dynamics and Kinematics

Sample Exam

Fall 2024

Formula sheet

$$x = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} \Leftrightarrow \tilde{x} = \begin{pmatrix} 0 & -x_3 & x_2 \\ x_3 & 0 & -x_1 \\ -x_2 & x_1 & 0 \end{pmatrix} \quad (1)$$

$$r = \frac{\omega \wedge v}{\|\omega\|^2}; \quad \lambda = \frac{\omega^\top v}{\|\omega\|^2} \quad (2)$$

$$\text{Ad}_{H_a^b} = \begin{pmatrix} R_a^b & 0 \\ \tilde{p}_a^b R_a^b & R_a^b \end{pmatrix}; \quad \text{ad}_{T_a^{c,b}} = \begin{pmatrix} \tilde{\omega}_a^{c,b} & 0 \\ \tilde{v}_a^{c,b} & \tilde{\omega}_a^{c,b} \end{pmatrix} \quad (3)$$

$$e^A := I + A + \frac{A^2}{2!} + \frac{A^3}{3!} + \dots \quad (4)$$

$$\theta \tilde{\omega} \in \mathfrak{so}(3), \quad \|\hat{\omega}\| = 1 \Rightarrow e^{\theta \tilde{\omega}} = I + \tilde{\omega} \sin \theta + \tilde{\omega}^2 (1 - \cos \theta) \quad (5)$$

$$e^{\tilde{r}} = \begin{pmatrix} e^{\tilde{\omega}} & \frac{1}{\|\omega\|^2} ((I - e^{\tilde{\omega}})(\omega \wedge v) + \omega^\top v \omega) \\ 0 & 1 \end{pmatrix} \quad (6)$$

$$\hat{T}_{\text{rot}} = \begin{pmatrix} \hat{\omega} \\ r \wedge \hat{\omega} \end{pmatrix}; \quad \hat{T}_{\text{tr}} = \begin{pmatrix} 0 \\ \hat{v} \end{pmatrix}; \quad \hat{T}_{\text{scr}} = \begin{pmatrix} \hat{\omega} \\ r \wedge \hat{\omega} + \lambda \hat{\omega} \end{pmatrix} \quad (7)$$

$$(\dot{\text{Ad}}_{H_b^a}) = \text{Ad}_{H_b^a} \text{ad}_{T_b^{b,a}} \quad (8)$$

$$\text{ad}_{T_\bullet^{b,\bullet}} = \text{Ad}_{H_a^b} \text{ad}_{T_\bullet^{a,\bullet}} \text{Ad}_{H_b^a} \quad (9)$$

$$(\dot{\mathcal{P}}^i)^\top = \text{ad}_{T_i^{i,0}}^\top (\mathcal{P}^i)^\top + (W^i)^\top \quad (10)$$

$$M(q)\ddot{q} + C(q, \dot{q})\dot{q} + G(q) = \tau^\top \quad \left(G(q) = \frac{\partial V}{\partial q} \right) \quad (11)$$

Part I (Test)

Read carefully. There is no limit in the number of right or wrong answers in a question. Wrong answers will be penalised.

1. Consider an n -dimensional real vector space $\mathcal{V}^n$. Tick the true propositions.
 - ☐ Every possible basis of $\mathcal{V}^n$ contains infinitely many elements.
 - ☐ The dimension of the Lie Algebra of $\mathcal{V}^n$ is $\frac{n(n-1)}{2}$.
 - ☒ Any vector $v \in \mathcal{V}^n$ possesses a unique representation (with respect to a basis $e_i \in \mathcal{V}^n$ ($i \in \{1, \dots, n\}$)) of n components $v^i \in \mathbb{R}$, namely the one such that $v = \sum_{i=1}^n v^i e_i$ holds.
 - ☒ The map between components of a vector $v \in \mathcal{V}^n$ expressed in two different basis is always invertible and has $\mathbb{R}^n$ as both domain and co-domain.
 - ☒ A change of basis is always a linear map.
 - ☐ The map $\phi : \mathcal{V}^n \rightarrow \mathbb{R}$ defined as $v \mapsto \phi(v) := 0 \quad \forall v \in \mathcal{V}^n$, is not a covector because the map is not linear.
 - ☒ The map ϕ defined above always defines the null vector of the covector space $(\mathcal{V}^n)^*$.
2. Consider the Lie group $\text{SO}(3)$. Tick the true answers.
 - ☐ It is an infinite dimensional manifold.
 - ☒ It is a finite-dimensional manifold.
 - ☐ It is both a manifold and an Abelian group (meaning that $\forall g, h \in \text{SO}(3)$, $g \bullet h = h \bullet g$, where $\bullet$ denotes the group operation).
 - ☒ A basis for its Lie algebra contains 3 elements.
 - ☒ It is both a manifold and a non-Abelian group.
 - ☒ An element of $\text{SO}(3)$ is a 3×3 matrix.
3. Consider a homogeneous matrix $H_a^b \in \text{SE}(3)$. What is true about it?
 - ☐ Represents a rigid body motion from frame Ψ_a to frame Ψ_b expressed in Ψ_a .
 - ☒ Represents a rigid body motion from frame Ψ_b to frame Ψ_a expressed in Ψ_b .
 - ☒ Represents a rigid body motion from frame Ψ_b to frame Ψ_a expressed in Ψ_a .
 - ☐ The inverse homogeneous matrix $H^{-1} = H^\top$.
 - ☐ Is an element of a 3-dimensional Lie group.
 - ☐ $\dot{H}_a^b$ being left-translated to the identity is a twist expressed in Ψ_b .
 - ☐ Homogeneous matrices can be composed using the chain rule since they are $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ -tensors.
4. Consider a 1-DoF kinematic pair connecting body a to body b . The joint state is described by the variable q .
 - ☒ Torque τ produces the mechanical power $\tau(\dot{q})$ (covector τ applied to vector $\dot{q}$).

- ☒ The unit twist $\hat{T}_b^{a,a}$ is always constant.
- ☒ The kinematic relation is given by $H_b^a(q) = e^{Aq} H_b^a(0)$, where A is the unit twist $\hat{T}_b^{a,a}$.
- ☒ Exponential map $\exp : \mathfrak{se}(3) \rightarrow \text{SE}(3)$ is a surjective map, but not injective since $\exists T_1, T_2 \in \mathfrak{se}(3), T_1 \neq T_2 : e^{T_1} = e^{T_2}$.
- ☐ $H_a^b = -H_b^a$

5. The Geometric Jacobian:

- ☐ does not depend on the configuration of the robot.
- ☒ contains the expression of the unit twists that characterise the kinematic pairs of the robot, expressed in the base frame Ψ_0 .
- ☐ has always a non-null determinant if the robot is redundant.
- ☒ may lose rank in some configurations.
- ☐ Is a $\binom{0}{2}$ -tensor.
- ☒ represents a linear transformation from $T_q \mathcal{Q}$ to $\mathfrak{se}(3)$.

6. Tick what is true about Brockett's formula.

- ☐ It solves the inverse kinematics problem for serial link manipulators.
- ☒ It is computationally convenient since the exponentials possess a closed form solution which depends only on the current joint positions q , and the unit twists of the joints and the homogeneous matrix of the final link in the reference configuration (for $q = 0$).
- ☒ It solves the direct kinematic problem for serial link manipulators.
- ☒ It produces a 4×4 matrix as a result, which represents an element of $\text{SE}(3)$.
- ☐ It produces a matrix as a result, whose last row contains only zeroes.

7. Consider the dynamics of a rigid body $(\dot{\mathcal{P}}^i)^\top = \text{ad}_{T_i^{i,0}}^\top(\mathcal{P}^i)^\top + (W^i)^\top$, also referred to as Lie-Poisson reduction. Tick the true answers.

- ☒ The superscript i refers to a frame which is attached to the rigid body, since the equation is in the body-fixed frame.
- ☒ The first three elements of the vector $\text{ad}_{T_i^{i,0}}^\top(\mathcal{P}^i)^\top$ represent rotational gyroscopic effects and take the form $(J\omega_i^{i,0}) \wedge \omega_i^{i,0}$ (where J is a diagonal 3×3 matrix) if i refers to the principal inertia frame.
- ☐ In order for the equation to be numerically solved, one needs the information on the current configuration $H_i^0(t)$ of the rigid body.
- ☒ W^i is an external wrench which acts on the rigid body.
- ☒ Is an equation on the space $\mathfrak{se}^*(3)$.
- ☐ Is valid only in an inertial frame, like Newton's law for a point mass $\dot{p} = F$.

8. Consider the inertia tensor $M(q)$ in the Euler-Lagrange equations of a robot. Tick the true answers.

- ☒ Is always a square matrix.
- ☐ Is invertible only if the robot is not in a singular configuration.
- ☒ Defines uniquely the so called Coriolis matrix $C(q, \dot{q})$.
- ☐ Is a $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ -tensor on the configuration manifold $\mathcal{Q}$.
- ☒ Is a $\begin{pmatrix} 0 \\ 2 \end{pmatrix}$ -tensor since $\frac{1}{2}\dot{q}^\top M(q)\dot{q}$ is the scalar kinetic energy in the system.
- ☐ Has zero determinant in some configurations, called singular configurations.

Part II (Open question)

Brockett's formula is used to calculate the end-effector configuration $H_n^0(q)$ for a serial robot, given the joint coordinates q . Derive the expression for Brockett's formula for a 3-d.o.f. manipulator starting from the chain rule for H matrices and a kinematic pair relation:

$$H_i^j(q_i) = e^{\tilde{T}_i^{j,j} q_i} H_i^j(0)$$

Explain each step in your derivation. Keep all the indices.

Matrix exponential property $Be^AB^{-1} = e^{BAB^{-1}}$ might be helpful.

Comment on each variable (q , $\tilde{T}$, $H(0)$) in the final expression. *What does the variable represent? What space does it belong to?*

Solution:

Unit twists in Brockett's formula are expressed in the inertial frame. To relate the twist in the kinematic pair relation in the inertial frame, we apply coordinate transformation from Ψ_0 to Ψ_j for twists in a matrix form:

$$\tilde{T}_i^{j,j} = H_0^j(0) \tilde{T}_i^{0,j} H_j^0(0)$$

The unit twist in the relation is in the reference configuration $q = 0$, therefore the H -matrices are also in the reference configuration $H(0)$. Henceforth, we will omit the (0) after homogeneous matrices for clarity.

Thus, the kinematic pair relation can be written as:

$$H_i^j(q_i) = e^{\tilde{T}_i^{j,j} q_i} H_i^j = e^{H_0^j \tilde{T}_i^{0,j} H_j^0 q_i} H_i^j = H_0^j e^{\tilde{T}_i^{0,j} q_i} H_j^0 H_i^j = H_0^j e^{\tilde{T}_i^{0,j} q_i} H_i^0$$

using the given matrix exponential property.

The chain rule can be used to express the configuration of the end-effector. Inserting the expression obtained above:

$$\begin{aligned} H_3^0(q) &= H_1^0(q_1) H_2^1(q_2) H_3^2(q_3) = \\ &= \cancel{H_0^0} e^{\tilde{T}_1^{0,0} q_1} \cancel{H_1^0} H_1^0 e^{\tilde{T}_2^{0,1} q_2} \cancel{H_2^0} H_2^0 e^{\tilde{T}_3^{0,2} q_3} \cancel{H_3^0} = \\ &= e^{\tilde{T}_1^{0,0} q_1} e^{\tilde{T}_2^{0,1} q_2} e^{\tilde{T}_3^{0,2} q_3} H_3^0(0) \end{aligned}$$

The resulting expression is Brockett's exponential formula. It provides a closed-form solution for the configuration $H_3^0(q)$ of the end-effector with respect to joint coordinates q_i given its configuration $H_3^0(q = 0)$ in the reference configuration and unit twists $\tilde{T}_i^{0,j}$ ($q = 0$) of each joint in matrix form in the reference configuration expressed in the inertial frame Ψ_0 .

A tuple of generalized coordinates q is an element of the configuration manifold $\mathcal{Q}$. Unit twists lie in Lie algebra $\mathfrak{se}(3)$. Homogeneous matrix $H_3^0(0)$ is an element of Lie group $SE(3)$.

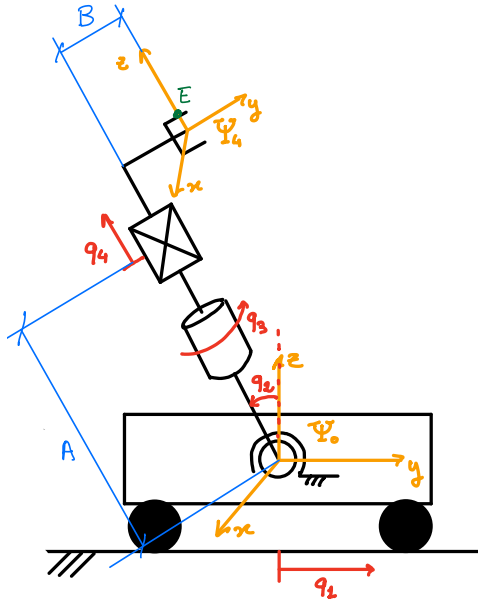
Part III (Exercise)

Consider the 4-DoF robot, the frames and distances represented in the figure. The robot consists of:

- A translational joint (q_1) with axis aligned in direction y of the base frame Ψ_0 ;
- A rotational joint (q_2) as in figure;
- A rotational joint (q_3) as in figure;
- A translational joint (q_4) allowing an extension of the vertical arm.

The point E is part of the end-effector and has constant coordinates $(0,0,1)^T$ in Ψ_4 , i.e. has homogeneous coordinates $E^4 = (0,0,1,1)^T$ in Ψ_4 . Make a choice for the reference configuration ($q_i = 0, i \in \{1, 2, 3, 4\}$) considering that the distance A in the direction of the vertical arm is a fixed constant which cannot be shrunk (i.e. the vertical arm has length $A + q_4$ with $q_4 \geq 0$).

1. Give Brockett's equation for direct kinematics and all unit twists in reference configurations that you need to calculate $H_4^0(q)$. *Note: There is no need to write out the matrix exponentials but you must calculate the vector products that appear inside your expressions.* (1 Point)
2. Compute the geometric Jacobian $J(q)$ such that $T_4^{0,0} = J(q)\dot{q}$. *Note: You must calculate the vector products that appear inside your expressions.* (2 Points)
3. Compute the velocity of the point E in the inertial frame Ψ_0 as function of q and $\dot{q}$. (0.5 Points)
4. Suppose an external wrench is applied on the end-effector with coordinates in Ψ_4 given by $W^4 = (0 \ 0 \ 0 \ 0 \ 1 \ 0)$, i.e. the wrench is constant in the end-effector frame. This could be the case if e.g. a propeller mounted on the end-effector would produce a constant thrust. Compute the generalised forces, as function of q and $\dot{q}$, which are experienced by the joints as effect caused by this wrench. (0.5 Points)



Part 3

$$1. H_q^0(q) = \prod_{i=1}^4 \left[\exp(\tilde{T}_i^{0,i-1}(q) q_i) \right] H_q^0(0)$$

$$\hat{T}_1^{0,0}(0) = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} \quad \hat{T}_2^{0,1}(0) = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} \quad \hat{T}_3^{0,2}(0) = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} \quad \hat{T}_4^{0,3}(0) = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$

$$H_q^0(0) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$2. T_q^{0,0} = J(q) \dot{q} \quad J(q) = \begin{pmatrix} \hat{T}_1^{0,0}(q) & \hat{T}_2^{0,1}(q) & \hat{T}_3^{0,2}(q) & \hat{T}_4^{0,3}(q) \end{pmatrix}$$

$$\hat{T}_1^{0,0}(q) = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} \quad \hat{T}_2^{0,1}(q) = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ -q_1 \end{pmatrix} \quad \hat{T}_3^{0,2}(q) = \begin{pmatrix} 0 \\ -\sin q_2 \\ \cos q_2 \\ q_1 \cos q_2 \\ 0 \\ 0 \end{pmatrix} \quad \hat{T}_4^{0,3}(q) = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ -\sin q_2 \\ \cos q_2 \end{pmatrix}$$

$$3. E^q = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 1 \end{pmatrix} \quad E^0 = H_q^0 E^q$$

$$\dot{E}^0(q) = \dot{H}_q^0 E^q + H_q^0 \dot{E}^q = \dot{H}_q^0 \underbrace{H_0^q}_{I} H_q^0 E^q = \tilde{T}_q^{0,0}(q) H_q^0(q) E^q \quad (\text{see 1,2})$$

$$4. W^{q,q} = (0 \ 0 \ 0 \ 0 \ 1 \ 0)$$

$$(W^{0,q})^T = \text{Ad}_{H_0^q(q)}^T (W^{q,q})^T$$

$$\tau^T = J^T(q) (W^{0,q})^T$$

$$\Rightarrow \tau = W^{q,q} \text{Ad}_{H_0^q(q)} J(q)$$

$$\text{, where } H_q^0(q) = (H_0^q(q))^{-1}$$

(see 1,2)